

SP-LT ADAPTIVE EMISSION

Mercatura Bootstrap and Long-Term Block Reward Algorithm

State-Preserving Logarithmic Time-Normalized Commodity Work-Yield Emission

Authoritative Phase 9 design • Technical rationale and consensus behavior

Executive summary

Mercatura does not use Bitcoin-style halvings, a fixed terminal supply, a treasury, or a programmed long-run inflation rate. Its emission law has two deliberately different phases: a deterministic 311,040-block launch bootstrap that creates exactly 20,000,000,000.00 MCA, followed by SP-LT, a slow state-preserving controller that adjusts future subsidy in response to the trend in proof-of-work supplied per unit of protocol subsidy.

The governing idea is commodity production rather than monetary targeting: accumulated stock is large and persistent; new MCA requires costly proof of work; and production changes slowly when mining economics persistently change. Gold is the conceptual reference for those structural properties, not a price peg, supply-ratio target, or literal template.

Final locked design. Bootstrap: heights 1–311,040, exactly 20B MCA, symmetric quintic, 23,782.34 MCA at both edges. Adaptive activation: height 311,041 at 23,782.34 MCA; first SP-LT-driven change at 311,042. SP-LT: 7-day / 180-day EMAs, $\kappa = 0.25$ (divisor 840,960), +25% / –33% annual log-rate limits, Q16.48 deterministic ln/exp, and a fixed 237.82 MCA actual-subsidy floor.

What the system is—and is not

- It is an immutable feedback law based only on prior consensus-observable chain state.
- It does not target USD, CPI, another asset, a fixed purchasing power, a target hashrate, a target annual inflation rate, or any numerical gold price / gold supply ratio.
- It does not guarantee real-world PoW security; the 237.82 MCA floor is a nominal miner-subsidy tail, not a security oracle.
- It keeps 100% of subsidy and transaction fees with PoW miners, but fees are excluded from the adaptive control signal.

1. Monetary philosophy and parameter map

SP-LT emerged from repeated attempts to preserve a commodity-like monetary objective without pretending that a fixed issuance schedule can know future mining economics. Gold provided the main conceptual reference: a large durable stock, costly marginal production, heterogeneous producers, and slow supply response. Mercatura copies those ideas only where a digital PoW system can do so coherently.

economic mining conditions → *PoW supplied* → *work yield* → *desired production capacity* → *actual MCA production*

1.1 Why there are no halvings or fixed supply cap

A halving schedule makes future quantity predictable by construction, but it also makes production insensitive to the amount of real work the market is willing to supply. Mercatura instead treats issuance as production of a monetary commodity. The protocol defines the production law, not a lifetime quantity. This gives up exact long-horizon supply forecasting in exchange for endogenous, slow-moving production. The permanent nominal floor prevents the adaptive process from mathematically reaching zero.

1.2 Current authoritative parameter set

Component	Locked value	Design function
Block target	150 seconds; 210,240 target blocks/year	Defines block-based time normalization
Bootstrap	311,040 blocks ≈ 540 target-days	Launch distribution + controller warm-up
Bootstrap stock	20,000,000,000.00 MCA exactly	Large two-decimal monetary stock
Edge subsidy	23,782.34 MCA	Launch start/end and adaptive handoff
Peak subsidy	104,818.49 MCA actual; 104,818.48304527 ideal	Middle of symmetric issuance hump
Work-yield signal	$z = \ln(W) - \ln(R_{base})$	Log work per unit protocol subsidy
Short EMA	4,032 blocks ≈ 7 days; $\alpha=2/4,033$	Fast trend state
Long EMA	103,680 blocks ≈ 180 days; $\alpha=2/103,681$	Slow baseline state
Controller	$q_{next} = q + (s-l)/840,960$	$\kappa = 0.25$ annualized desired-state gain
Actuator limits	+25% / -33% per year in log space	Limits actual reward velocity
Floor	237.82 MCA/block fixed nominally	Permanent actual-subsidy tail
Math	Signed Q16.48 state; integer ln/exp	Cross-platform consensus determinism

1.3 The acronym

SP-LT-CWYE originally meant State-Preserving Logarithmic Time-Normalized Commodity Work-Yield Emission. In current Mercatura documentation, it is normally shortened to SP-LT. “State-preserving” is the essential architectural distinction: desired production q is not discarded when the actual reward r is rate-limited or held above the floor.

2. Why gold is the conceptual reference

Mercatura is not attempting to reproduce gold ounce-for-ounce or copy today's gold stock-to-flow ratio. The gold analogy is structural: a large accumulated stock coexists with costly marginal production, producers have different costs, and production capacity responds to economics with inertia rather than by a predetermined halving clock.

2.1 Large durable stock versus marginal new production

Gold's monetary character is strongly influenced by the fact that the above-ground stock is very large relative to annual mine production. World Gold Council end-2025 estimates put above-ground gold at roughly 220,700 tonnes, while 2025 mine production was about 3,671.6 tonnes. As a simple scale comparison, annual mine flow was therefore about 1.66% of the existing above-ground stock.

$$3,671.6 \text{ t} / 220,700 \text{ t} \approx 1.66\%$$

Mercatura begins with no inherited centuries-old stock, so the 20B-MCA bootstrap is a one-time initialization device: it creates a large distributed stock before long-run adaptive production takes over. That is an engineered starting condition, not something gold itself ever experienced.

2.2 Costly and heterogeneous marginal producers

Gold mines differ in ore grade, energy use, labor, capital intensity, jurisdiction, and processing cost. Market prices and input costs change which deposits or portions of an ore body are economical. Mercatura seeks an analogous margin through heterogeneous MercaHash miners: hardware efficiency, electricity price, cooling, capital cost, and alternative mining opportunities differ across operators.

That is the economic motivation for work yield. SP-LT does not ask for a gold price or an MCA price. It observes how much consensus work competes for each unit of protocol subsidy. Persistently more work per unit of new MCA is treated as evidence that a higher production state may be economically supportable; persistently less work points the other way.

$$z_h = \ln(W_h / R_{h,base})$$

2.3 Production inertia: the reason for q and r

Gold mine supply is famously slow to respond. New projects require exploration, permitting, financing, construction, and commissioning; the World Gold Council notes that it can typically take 10–20 years after discovery before a mine is producing refinable material. Existing operations can adjust sooner, for example by changing which ore grades are economic, but aggregate mine supply still does not move instantly with price.

SP-LT models the same idea abstractly. q is the production level current evidence says would eventually be justified; r is the production level the protocol is presently allowing. The rate limiter prevents actual subsidy from instantly jumping to q. State preservation then keeps q intact while r catches up, just as an economically justified expansion can exist before physical productive capacity is built.

Structural analogy, not numerical imitation. Mercatura's activation subsidy annualizes to roughly 25% of its 20B bootstrap stock—far above gold's present mine-flow/above-ground-stock ratio. That is intentional launch formation, not a claim that Mercatura already has gold-like mature issuance. The analogy concerns the production mechanism and the possibility of later maturation, not matching gold's current percentage.

3. Where the gold analogy ends

The analogy is useful only if its limits are stated explicitly. Gold is a physical element with geology, recycling and non-monetary demand. Mercatura is a digital consensus system whose miners can reallocate compute rapidly and whose monetary law is executable code. These are not minor details; they are the main reasons SP-LT must be treated as an experiment in commodity-like behavior rather than a claim of equivalence.

Dimension	Gold	Mercatura / implication
Scarcity source	Physical geology and finite ore bodies.	Consensus-defined issuance; no natural ore body or geological reserve.
Production inputs	Ore grade, land, labor, capex, energy, processing and regulation.	Hardware, electricity, cooling, capital and opportunity cost.
Response speed	New mines commonly take many years; supply responds slowly.	Reward changes are algorithmic; miners can enter, leave or switch chains much faster.
Depletion	Deposits can be exhausted; grades and reserve economics change.	No natural depletion. Mercatura deliberately avoids inventing synthetic digital geology.
Recycling	A major price-responsive supply source; most mined gold remains recoverable.	Transfers do not create new MCA. There is no monetary-supply analogue to recycling.
Demand uses	Jewellery, investment, official reserves and technology.	Digital monetary/network demand; no physical industrial or jewellery use.
Security	Gold mining produces metal but does not secure a transaction ledger.	PoW miners both receive issuance and secure Mercatura consensus.
Producer mobility	A gold mine cannot become a copper mine in minutes.	CPU miners may redirect compute among compatible opportunities quickly.
Observability	Costs and reserves are dispersed, estimated and firm-specific.	W and R are exact on-chain quantities, but the economic cause of a W/R change is ambiguous.

3.1 No synthetic geology

Mercatura does not add an artificial ore-grade decline, reserve curve, or programmed depletion schedule merely to look more like gold. Such a curve would be invented by protocol designers rather than discovered by the network. MercaHash already imposes real resource costs, so SP-LT tries to react to observed production economics instead of fabricating a digital geology.

3.2 The largest break in the analogy: miner switching

Gold production is geographically and physically sticky. CPU mining is not. If comparable proof-of-work opportunities exist, miners can move capacity toward whichever chain is temporarily more profitable. In that world, W/R may partly measure short-run arbitrage and competitor-chain conditions rather than durable MCA economics. This is the central empirical uncertainty that the controller mathematics cannot eliminate.

3.3 A fair statement of the design objective

- Emulate a large-stock / marginal-flow monetary structure, not gold's exact issuance percentage.
- Require real resource expenditure for new units, while recognizing that compute is more mobile than mines.
- Let persistent economic conditions influence production, but make actual production adjust slowly.
- Avoid discretionary monetary intervention: the response law is fixed in consensus even though future quantity is not.

What SP-LT does not promise. It does not make MCA price-stable, guarantee increasing stock-to-flow, guarantee PoW security, identify the cause of every mining change, or prove that digital miners will behave like gold producers. Gold is the design reference for selected monetary-production properties, not evidence that the two systems are economically identical.

Gold-market context used here: World Gold Council, Gold Market Primer: Market size and structure (Aug. 2026); Gold Demand Trends: Full Year 2025 — Supply (Jan. 2026); How Gold Is Mined: The Lifecycle of a Gold Mine; and Gold Supply. End-2025 / 2025 figures are contextual comparisons only and are not SP-LT consensus inputs.

4. Bootstrap phase: exact stock, fairer timing, smooth handoff

4.1 Why a bootstrap exists at all

SP-LT needs both an initial monetary stock and a meaningful work-yield history. Starting directly with a tiny reward would leave a two-decimal currency with poor denomination granularity and a thinly distributed initial stock; starting with a huge short-lived reward would concentrate issuance in the earliest miners and create a severe reward cliff. The bootstrap therefore solves three launch problems together: stock formation, distribution over time, and a smooth transition into adaptive production.

4.2 Why 20 billion MCA

Mercatura has only 100 base units per MCA. Earlier denomination research found that small nominal stocks make the 0.01 MCA atomic unit economically coarse long before reward arithmetic itself becomes difficult. That research identified 5–20B MCA as an acceptable range and emphasized that a larger nominal stock does not by itself make the currency less scarce: relative issuance, not the printed coin count, determines dilution. The final consensus choice was 20B MCA, the upper end of that researched range.

Historical-record note. The preserved research documents explain why Mercatura moved to a multi-billion-unit stock and why 5–20B was considered acceptable. They do not preserve one single definitive argument for choosing exactly 20B rather than the earlier 10B candidate. The 20B value is a later explicitly locked consensus decision and supersedes the earlier research candidate.

4.3 Why 311,040 blocks / about 540 target-days

At 150-second target spacing there are 576 blocks/day. 540 target-days therefore gives exactly 311,040 rewarded blocks. Earlier launch simulations showed that extending a smooth distribution from one year toward roughly 18 months materially reduced first-miner concentration, while extending from 18 months to roughly two years produced much smaller incremental benefit. The 540-day choice is therefore a practical fair-distribution / time-to-adaptive-production compromise, not a magical economic constant.

$$576 \text{ blocks/day} \times 540 \text{ days} = 311,040 \text{ bootstrap blocks}$$

4.4 Symmetric quintic smoothstep reward shape

The bootstrap is split into two equal halves of 155,520 blocks. Within each half, the normalized reward offset follows the classic quintic smoothstep $Q(x)$. It rises from zero offset to full amplitude in the first half and mirrors back down in the second.

$$Q(x) = 6x^5 - 15x^4 + 10x^3, \quad 0 \leq x \leq 1$$

$$f_h = Q((h-1)/(N-1)) \text{ for } 1 \leq h \leq N; \quad f_h = Q((H-h)/(N-1)) \text{ for } N+1 \leq h \leq H$$

Q was chosen because $Q(0)=0$, $Q(1)=1$, and both its first and second derivatives vanish at the endpoints. That avoids an abrupt slope or curvature change at launch, at the center peak, and at the return to the edge reward. Symmetry avoids deliberately front-loading issuance to the earliest miners and guarantees the final bootstrap reward returns to the same level used for adaptive activation.

4.5 Exact amplitude and reward levels

$$\tilde{R}(h) = R_0 + A \cdot f_h, \quad R_0 = 2,378,234 \text{ base units}, \quad A = 1,969,178,276 / 243 \text{ base units}$$

This yields an ideal center reward of 104,818.48304527 MCA before cent quantization. The actual two center blocks are 104,818.49 MCA. The average bootstrap subsidy is 64,300.41152263 MCA/block because the total is fixed at 20B MCA.

5. Bootstrap quantization, distribution, and activation handoff

5.1 Why rewards are quantized cumulatively

Mercatura cannot pay fractional base units. Rounding each ideal block reward independently would accumulate drift and could break the exact 20B target or the intended mirror symmetry. Instead, the protocol quantizes cumulative ideal issuance and defines each block reward as the difference between two adjacent cumulative integers.

$$I_h = \sum_{j=1 \dots h} \tilde{R}(j); \quad C_h = \lfloor I_h \rfloor; \quad R(h) = C_h - C_{h-1}$$

$$\sum_{h=1 \dots 311040} R(h) = 2,000,000,000,000 \text{ base units} = 20,000,000,000.00 \text{ MCA exactly}$$

The implementation evaluates the first-half cumulative polynomial in $O(1)$ time using exact integer power sums, then defines the second half by mirroring the already-quantized first-half rewards. That preserves exact cent-level symmetry. Only the final rational cumulative issuance is floored; ideal individual rewards are never independently rounded.

5.2 What the launch distribution actually looks like

Target time	Cumulative issued	Share of 20B bootstrap
30 days	415,144,015.36 MCA	2.075720%
90 days	1,483,543,861.63 MCA	7.417719%
180 days	4,816,877,194.97 MCA	24.084386%
270 days / midpoint	10,000,000,000.00 MCA	50.000000%
540 days / end	20,000,000,000.00 MCA	100.000000%

Because cumulative cent quantization is exact rather than cosmetic, very early blocks can show tiny ± 0.01 MCA microstructure even while the underlying ideal curve is monotonic. For example, the first visible one-cent increase occurs around height 118; a later block can momentarily return to the prior cent because the protocol is tracking cumulative ideal issuance, not rounding each block in isolation.

5.3 The 311,040 → 311,041 → 311,042 handoff

Height	Subsidy rule	Controller meaning
311,040	23,782.34 MCA	Final bootstrap block
311,041	23,782.34 MCA	First adaptive-era block; q and r initialized to $\ln(2,378,234 \text{ base units})$
311,042	First possible SP-LT-driven change	Uses state observed through block 311,041

The equal boundary reward removes a launch-to-adaptive reward cliff. More importantly, the controller does not accumulate an adaptive command backlog during bootstrap. Only the short and long work-yield EMAs are warmed. At activation, q and r are set explicitly to the edge subsidy; the warmed EMA states carry across. This separates “learn the work-yield baseline” from “start changing money production.”

5.4 Why the edge reward is 23,782.34 MCA

At 210,240 target blocks/year, the edge reward annualizes to approximately 4,999,999,161.60 MCA/year, essentially 25% of the 20B activation stock. This is an initialization production rate, not a permanent inflation target. SP-LT is free to move production thereafter.

$$23,782.34 \times 210,240 / 20,000,000,000 \approx 25\% \text{ annualized at activation}$$

6. SP-LT signal and dual-timescale filtering

6.1 Work yield: what the controller observes

For each accepted block h , Mercatura computes W_h using Bitcoin Core-style GetBlockProof. This is deterministic consensus work derived from the block target—not displayed difficulty, wall-clock hashrate estimates, or an external oracle. The protocol subsidy R_h is measured in integer base units.

$$z_h = \ln(W_h) - \ln(R_h, \text{base}) = \ln(W_h / R_h, \text{base})$$

The logarithm is fundamental. A multiplicative change becomes an additive difference, making the signal scale-independent: doubling work and doubling subsidy together leaves work yield unchanged. Positive z means more work is competing per unit of newly produced MCA; negative movement means less work per unit.

6.2 Why GetBlockProof rather than estimated hashrate

- It is already a consensus-native integer representation of work implied by the target.
- It does not require timestamp-based hashrate estimation, which would add noise and manipulation surface.
- It is branch-local: a reorg naturally supplies a different work history and therefore a different emission state.
- It can be recomputed exactly during startup/reindex from block ancestry and targets.

6.3 Why transaction fees and miner underclaim are excluded

The control denominator is the protocol subsidy commanded by consensus. Transaction fees are volatile, depend on transaction demand, and are not a quantity SP-LT controls. Including fees would let unrelated fee spikes alter the monetary controller. Likewise, a miner must not be able to change future monetary state by deliberately underclaiming a valid subsidy. The implementation therefore feeds protocol subsidy—not actual coinbase take—into z .

6.4 Two exponential moving averages

$$s_h = s_{h-1} + 2(z_h - s_{h-1})/4,033$$

$$l_h = l_{h-1} + 2(z_h - l_{h-1})/103,681$$

$$e_h = s_h - l_h$$

The 4,032-block span is approximately seven target-days: long enough to smooth block-to-block and daily noise, but short enough to detect persistent changes on a human-scale week. The 103,680-block span is approximately 180 target-days: it is the slow economic baseline against which the weekly state is compared.

Earlier parameter studies compared 90-day, 180-day, 365-day and longer baselines. Very long baselines filtered more noise but created excessive integrated controller gain and slower normalization; shorter baselines reduced the distinction between temporary and structural changes. The 7d/180d pair was the recurring compromise and was retained in the final design.

No absolute setpoint. SP-LT does not say what W/R “should” be. It responds to the short-versus-long trend of its own history. That avoids a hardcoded target hashrate or external price oracle, but it also means the work-yield signal remains economically underidentified: work can move because of price, electricity, hardware, competitor chains, rental markets, or strategy.

7. State preservation: desired production q versus actual production r

7.1 Desired state

$$q_{h+1} = q_h + e_h / 840,960$$

Because $840,960 = 4 \times 210,240$, the implemented divisor corresponds to $\kappa = 0.25$ per target-year of sustained log error. q is stored in logarithmic subsidy space. It represents the production level the work-yield trend is asking for, independent of how quickly the actual reward is allowed to move.

The architecture intentionally has no dead zone. A dead zone can prevent equal-and-opposite small trends from cancelling, creating hysteresis or a ratchet. With no dead zone, the controller keeps recording even weak evidence, while the very slow gain and long EMA limit short-term influence.

7.2 Actual rate-limited state

$$r_{h+1} = r_h + \text{clip}(q_{h+1} - r_h, d_-, d_+)$$

$$d_+ = \ln(1.25)/210,240; \quad d_- = \ln(0.67)/210,240$$

The actual log-reward state r can expand by at most 25% per target-year or contract by at most 33% per target-year. The limit is block-based rather than timestamp-based, so miners cannot accelerate or slow monetary adjustment by manipulating timestamps. Log-space limits make the cap multiplicative and scale-free: the same percentage rule applies whether the subsidy is hundreds or tens of thousands of MCA.

The asymmetry is deliberate. Faster contraction is economically attractive when work yield deteriorates, but removing subsidy too quickly can be security-procyclical because poor mining economics are exactly when miners are leaving. Earlier stress studies retained +25%/–33% as the compromise between production responsiveness and avoiding abrupt subsidy withdrawal.

7.3 Why q must remain separate from r

This is the meaning of state-preserving. If a large persistent change pushes the desired reward faster than r is allowed to move, q continues recording the full desired adjustment. When the short/long EMA differential later decays, r can continue catching up to the preserved target. A directly clipped controller instead loses part of the intended equilibrium while saturated because its input disappears before the actuator catches up.

Key invariant idea. The velocity limit constrains actual production capacity; it must not rewrite the economic state that justified that capacity. Destroying q during saturation converts a physical-speed limit into a lossy monetary-policy rule.

7.4 Parameter evolution: $\kappa = 0.25$

Earlier adversarial research explored lower gains, especially $\kappa = 0.05$ and 0.10 , and warned that high gain can widen long-horizon monetary tails when miner reward elasticity is large. The final Mercatura revision later locked $\kappa = 0.25$ / divisor $840,960$. That later value is authoritative for consensus. The archived material available for this document preserves the lock but not a single detailed written derivation explaining the increase from the older research preference, so the distinction is recorded explicitly rather than retroactively inventing a rationale.

8. Permanent subsidy floor and deterministic consensus arithmetic

8.1 Fixed nominal tail: 237.82 MCA

$$R_{actual}(h) = \max(R_{candidate}(h), 23,782 \text{ base units})$$

The floor is 237.82 MCA forever unless consensus changes. At the 20B activation stock it represents about 0.25% annual nominal issuance (49,999,276.80 MCA/year) and exactly 1% of the 23,782.34 MCA activation reward. It is fixed in MCA rather than recalculated as a percentage of future stock, so its proportional dilution naturally declines as supply accumulates.

$$237.82 \times 210,240 / 20,000,000,000 \approx 0.25\% \text{ annualized at activation}$$

A fixed tail was chosen over a percentage floor or stock-scaled square-root floor because those alternatives add continuing monetary expansion without actually observing real attack cost. A larger subsidy can support miners, but no deterministic coin-supply formula can guarantee security because price, hardware, energy, liquidity, miner concentration, rentals, and competing opportunities remain external.

8.2 Floor actual subsidy only—never q

When r implies less than 237.82 MCA, only the paid subsidy is raised to the floor. q and r remain untouched. This can leave a hidden desired state below the floor for a long time, and that is intentional. Extreme collapse/recovery simulations found that clamping or resetting q destroys the stored negative history and can create severe post-recovery overshoot or trap the controller at the floor. The floor is therefore an actuator output constraint, not a controller-state constraint.

8.3 Why Q16.48 and integer-only ln/exp

Consensus nodes must agree bit-for-bit. Native floating point is unsuitable because compiler, architecture, library, and optimization differences can change rounding at boundaries. Mercatura stores z , s , l , e , q and r as signed Q16.48: an int64 value scaled by 2^{48} . This provides roughly 3.55×10^{-15} resolution while retaining ample log-domain range.

$$real_value = q48_integer / 2^{48}$$

The implementation uses wider Q160.96 intermediates for ln/exp, arith_uint256 for bounded products, deterministic range reduction, a fixed atanh-series through odd power 95 for ln, and a fixed 40-term series for exp on a reduced interval. Irrational constants are pre-quantized; there is no platform-dependent math library call.

8.4 Explicit rounding rules

Operation	Consensus rule	Reason
Bootstrap	Floor cumulative rational issuance	Exact total + mirrored cent-level symmetry
EMA/controller division	Signed division truncates toward zero	Deterministic symmetric integer rule
Q96 → Q48	Nearest; exact half upward for positive magnitude	Permanent fixed-point conversion rule
exp → CAmount	Positive nearest; exact half upward	Deterministic base-unit subsidy
Irrational constants	Nearest; exact half away from zero	Fixed consensus constants

8.5 Reorgs, restart, and branch-local state

Emission state is chain state. Each indexed block has the EMA/controller state implied by its own ancestry. Disconnecting a block restores the exact parent state; a replacement branch recomputes deterministically from that branch's work and protocol subsidies. Startup and reindex reconstruct the same state from ancestry rather than relying on mutable global history. This is essential because two forks with the same height can legitimately have different work histories and therefore different next subsidies.

9. End-to-end algorithm, rejected alternatives, and design limits

9.1 Consensus update order

Step	For block h / child h+1
1	Derive the protocol subsidy command for block h solely from the parent state.
2	Mine/validate block h and obtain deterministic GetBlockProof work W_h .
3	Compute $z_h = \ln(W_h) - \ln(\text{protocol subsidy } R_h)$. Fees and miner underclaim do not enter.
4	Update short and long EMA states with z_h . During bootstrap, stop here.
5	From height 311,041 onward, compute $e_h = s_h - l_h$ from the accepted parent state.
6	Advance desired q by $e_h/840,960$; do not clip q.
7	Move actual r toward q by the per-block +25%/–33% log limits.
8	Exponentiate r, round to base units, then apply the fixed 23,782-base-unit paid-subsidy floor.
9	Store the resulting branch-local state so the next child command is deterministic and reorg-safe.

9.2 Alternatives examined and why they were not retained

Alternative	Problem identified
30-day / heavily front-loaded bootstrap	Concentrates issuance in earliest miners and creates an abrupt post-launch production change.
Constant or declining launch reward	Constant shape mismatches adaptive entry unless specially calibrated; declining schedules intentionally favor earliest miners.
Very long 730-day bootstrap	Only small distribution improvement over 540 days while delaying adaptive production materially.
Displayed difficulty / estimated hashrate signal	Adds representation/timestamp estimation issues; GetBlockProof is cleaner consensus work.
Fees inside W/R denominator	Lets fee-market volatility contaminate monetary policy; not controlled by SP-LT.
Dead zone around $e=0$	Can prevent small opposite signals from cancelling and create hysteresis/ratchets.
Clip desired q directly	Loses equilibrium information during saturation—the defect SP was created to fix.
Floor or reset q	Extreme recovery tests produced pathological overshoot or floor trapping.
Percentage or stock-scaled floor	Adds perpetual dilution without a genuine security oracle.
Ordinary reward upside cap	Would create another saturation surface and contradict open-ended commodity production.
Timestamp-normalized actuator speed	Would expose monetary adjustment to timestamp behavior; block counts are deterministic.

9.3 What SP-LT can infer—and what it cannot

Work yield is coherent as a commodity-production signal: if MCA becomes more valuable, energy becomes cheaper, or hardware becomes more efficient, more real work can economically compete for each unit of new MCA. But one scalar W/R observation cannot identify which cause moved. Compatible miners may also switch among chains when relative profitability changes. SP-LT therefore fixes a control/actuator problem; it does not solve the economic-sensor identification problem.

9.4 The central trade

Mercatura deliberately trades exact future-supply predictability for an immutable production law that can adapt slowly to persistent proof-of-work economics. The bootstrap is fully predetermined; the mature subsidy is not. The design is credible only if users can verify that every adjustment is the mechanical result of prior chain state, with no governance switch, discretionary committee, oracle, or hidden parameter.

Authoritative versus historical material. The exact consensus values in this document are the current Phase 9 values: 20B bootstrap stock, 23,782.34 MCA edge, $\kappa=0.25$, 237.82 MCA floor, Q16.48, and the 311,041/311,042 activation sequence. Earlier research documents contain superseded candidates such as 10B stock, $\kappa=0.05/0.10$, alternative floor functions, and different activation initialization proposals. Those older studies are used here only to explain the reasoning process and rejected alternatives.

10. Compact reference

Bootstrap: $R(h) = [\Sigma_{j \leq h} \tilde{R}(j)] - [\Sigma_{j \leq h-1} \tilde{R}(j)]$, $\Sigma R(h) = 20,000,000,000.00$ MCA

Signal: $z_h = \ln(W_h) - \ln(R_h, \text{base})$; EMAs: $\alpha_s = 2/4,033$, $\alpha_l = 2/103,681$

Desired: $q_{h+1} = q_h + (s_h - l_h)/840,960$

Actual: $r_{h+1} = r_h + \text{clip}(q_{h+1} - r_h, \ln(0.67)/210,240, \ln(1.25)/210,240)$

Subsidy: $R_{h+1} = \max(\text{round}(\exp(r_{h+1})), 23,782 \text{ base units})$

Technical basis: current Mercatura roadmap v1.4, implemented Phase 9 consensus sources, archived SP-LT/bootstrap research, and the World Gold Council sources identified in Section 3 for contextual gold-market facts. Current locked Phase 9 values supersede earlier research candidates.